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AgriSureGIS: A WebGIS-Based Typhoon Parametric Insurance Assessment for PCIC Region X

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Geographic Information Systems (GIS) are widely used around the world to manage and analyze location-based data. In agriculture, GIS helps monitor crops, measure land area, assess environmental risks, and support decision-making. A recent systematic review by Raihan (2024) explains that GIS improves agricultural planning, sustainability, and data-driven decision-making by combining spatial and non-spatial information. GIS allows agricultural agencies to transition from basic record-keeping to advanced spatial analysis, enabling them to visualize farmland distributions and assess environmental vulnerabilities with high precision.

In regions highly susceptible to extreme weather events, such as the Philippines, agricultural insurance plays a critical role in managing climate risks and ensuring the economic survival of small-scale farmers. However, traditional damage assessment relies heavily on slow, manual field verification. To overcome this, modern agricultural insurance is shifting toward data-driven, index-based parametric models that utilize real-time meteorological data such as typhoon wind velocities and rainfall indices to automate hazard detection and accelerate claim processing (Nguyen-Huy et al., 2024).

Despite these technological advancements, the Philippine Crop Insurance Corporation (PCIC) Region X faces significant operational bottlenecks due to the persistence of fragmented "data silos." Currently, based on the conducted interview, the agency manages over 500,000+ farmer records using disconnected CSV files, standalone GPX coordinates, and isolated desktop GIS software. This lack of a centralized infrastructure increases manual workloads and hinders rapid, multi-user damage assessments during typhoon emergencies (Abu-ElSondos, 2025). Literature suggests that unified Web-GIS frameworks significantly improve data integration, accessibility, and transparency for insurance systems (Bhakal et al., 2024; Zhang et al., 2024; Rabuh et al., 2024).

The need for modernization is urgent, as PCIC's records are projected to grow significantly over the next five years. While software like QGIS provides powerful analysis, it lacks a customized, web-based interface tailored to PCIC's specific workflow. Furthermore,

moving toward unified systems can eliminate the organizational "silos" that slow down risk assessment and service delivery (Alshareef et al., 2024). Integrating spatial and satellite-based data is now essential for supporting accurate, high-speed insurance programs (Nguyen et al., 2025).

Data silos occur when different parts of an organization store and manage data separately, making it difficult to share information across departments. This can slow decision-making, reduce operational efficiency, and increase costs in organizations, including insurance companies. Alshareef et al., (2024) explain that in the insurance sector, organizational culture and leadership play a significant role in eliminating data silos and adopting business intelligence systems that support integrated data management. Their study in Jordan's insurance sector highlights that modern insurance organizations benefit when data systems are unified, which leads to faster access to information, better risk assessment, and improved service delivery.

Given the global advancements in Web-GIS and geospatial insurance systems and the need to eliminate data silos in insurance data management, there is a clear opportunity to modernize PCIC's data system. Therefore, this study proposes a localized Web-GIS platform for PCIC Region X. The system will centralize farmer profiles and insurance records in a secure, localized spatial database to ensure data privacy (Ilba, 2021). Crucially, the platform will feature an automated disaster response module that extracts real-time weather data from Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) Tropical Cyclone Bulletins. By overlaying typhoon wind signals against the PCIC "Risk Exposure List," the system will rapidly calculate affected farm polygons and potential claims (Chen et al., 2025). Furthermore, the system will support spatial-to-attribute data linking, advanced filtering tools, mapping visualization, and automated PDF report generation. By developing and pilot-testing this localized, data-driven system, the study aims to improve operational efficiency, secure data privacy, and accelerate disaster response in crop insurance management in the Philippines.

1.2. Statement of the Problem

1.2.1 General Problem

The PCIC Region X manages a vast inventory of geospatial data, including farm boundaries (polygons) and agricultural insurance records containing personal information of farmers. However, this data is currently "siloed" within localized files and standalone QGIS software. In the landscape of modern agricultural insurance, this fragmented data architecture creates severe operational bottlenecks. When extreme weather events occur, GIS specialists must manually extract data from PAGASA Tropical Cyclone Bulletins and individually cross-reference typhoon wind signals against localized farm polygons to estimate crop damages. This manual processing is highly inefficient, prone to human error, and causes significant delays in the parametric insurance indemnification process. Furthermore, the agency lacks a centralized spatial platform capable of rapidly filtering risk exposures and generating immediate damage assessment reports.

1.2.2 Specific Problem

Specifically, the study seeks to address the following problems:

- The lack of a centralized spatial database to manage the growing volume of farmer metadata alongside their respective farm lot polygons.
- The absence of an interactive web-based dashboard that allows personnel to rapidly execute spatial filtering of farmer records by municipality, province, or RSBSA ID.
- The inefficiency of manually downloading and extracting real-time weather data and wind signal thresholds from public PAGASA Tropical Cyclone Bulletins.
- The inability to automatically cross-reference typhoon wind signals against farm locations to instantly estimate potential parametric insurance claims based on crop stage and risk exposure.
- The lack of an automated reporting tool to instantly generate and export spatial damage assessment summaries for management review and payout approval.

1.3. Objectives of the Study

1.3.1 General Objective

The general objective of this study is to develop and implement a localized WebGIS platform for the Philippine Crop Insurance Corporation (PCIC) Region X that centralizes spatial farm boundaries and farmer insurance metadata. This system aims to eliminate fragmented data silos, automate the extraction of weather data from PAGASA, and streamline parametric disaster risk assessments to expedite insurance claim settlements.

1.3.2 Specific Objectives

Specifically, this study aims to:

- a) To **design and develop** a centralized spatial database (using PostgreSQL/PostGIS) to efficiently store, manage, and link the growing volume of farmer metadata with their respective farm lot polygons.
- b) To **develop an interactive web-based mapping dashboard** equipped with spatial filtering tools, allowing personnel to instantly isolate and view farmer records by municipality, province, or RSBSA ID.
- c) To **create an automated data extraction module** capable of parsing real-time weather data and typhoon wind signal thresholds directly from public PAGASA Tropical Cyclone Bulletins.
- d) To **build a spatial processing engine** that automatically overlays extracted typhoon wind signals against farm locations to instantly calculate risk exposure and estimate parametric insurance claims based on crop stage.
- e) To **integrate an automated reporting feature** that allows users to instantly generate and export spatial damage assessment summaries into CSV format for management review and payout approval.
- f) To **evaluate the functionality and usability** of the system by conducting system testing and usability testing with an authorized PCIC personnel, ensuring the tool effectively meets their operational requirements for rapid disaster risk assessment.

1.4. Scope and Delimitation

1.4.1 Scope

The development of this AgriSureGIS platform is specifically designed to modernize the spatial data management and insurance verification workflows of the Philippine Crop Insurance Corporation (PCIC) Regional Office X. The scope of the project encompasses the following core components:

This study focuses on developing a centralized, web-based platform that integrates farmer profiles, insurance metadata, and spatial farm boundaries into a secure, localized spatial database (such as PostGIS) rather than a public cloud repository. At its core, the system features an automated disaster response module with a dedicated geoprocessing tool designed to extract real-time weather data from PAGASA Tropical Cyclone Bulletins, overlay typhoon wind signals against the PCIC's Risk Exposure List, and automatically calculate the duration of risk exposure to estimate potential claims.

To support these workflows, the platform delivers essential GIS functionalities, including seamless spatial-to-attribute data linking and an interactive mapping dashboard equipped with advanced filtering tools to sort data by farmer name, municipality, province, barangay, or RSBSA ID. Furthermore, the study incorporates satellite-based crop verification by utilizing Google Earth Engine and Synthetic Aperture Radar (SAR) imagery to verify whether an area is actively planted within a farmer's boundaries, effectively bypassing the need for post-typhoon physical farm visits. Structurally, this technological infrastructure and its primary dependencies rely on open-source repositories like GitHub for custom parsers and external API keys like GEE for base mapping, meaning the platform's mapping and data extraction capabilities are intentionally aligned with the availability, rate limits, and continuous support of these external web services.

1.4.2 Delimitations

To ensure the feasibility and successful deployment of the system within the given timeframe, the study is bound by the following delimitations:

This study is subject to several technical limitations, primarily that AgriSureGIS operates strictly as a standalone WebGIS platform without direct API integration into the PCIC's centralized system, relying instead on manual data imports from the legacy PCIC Automated Business System via standardized CSV and GPX files. Furthermore, platform accessibility is confined to an office-based environment for authorized GIS specialists and management using desktop or laptop computers; the system lacks a mobile application and does not support offline field data collection or offline-to-online synchronization for field adjusters.

In terms of data processing and analysis, the automated weather extraction module is specifically tailored to the standardized PDF formats of PAGASA Tropical Cyclone Bulletins rather than live API feeds from global meteorological agencies. Additionally, while the platform streamlines daily insurance workflows and rapid damage assessments, it is dependent on external tools and is not intended to replace complex spatial analysis features found in professional desktop GIS software like QGIS or Google Earth Engine. Finally, regarding parametric accuracy and basis risk, it is beyond the scope of this IT Capstone Project to evaluate or improve the meteorological or actuarial accuracy of the parametric index itself; the platform operates under the assumption that the predefined Risk Exposure matrices provided by the PCIC and the weather data from PAGASA are inherently accurate.

1.5. Significance of the Study

To the PCIC, this study provides a centralized and secure WebGIS platform that integrates farmer profiles, insurance records, and farm boundaries into one system. Instead of using separate CSV files, GPX files, and desktop GIS software, PCIC can manage all data in a single web-based platform. Furthermore, the introduction of the parametric assessment tool drastically reduces the manual effort and time required to calculate risk exposure during severe weather events.

To the GIS specialists, this system delivers an advanced, centralized desktop tool that eliminates tedious manual data processing. The interactive dashboard and spatial-to-attribute

linking allow them to instantly visualize and verify farm records. Furthermore, the automated PAGASA data extraction drastically reduces the manual workload required during typhoon damage assessments, saving significant time and minimizing human error.

To the farmers, this system delivers a vastly accelerated damage verification and indemnification process. Because the platform utilizes meteorological indices to automate assessments rather than relying entirely on lengthy manual field inspections, farmers will experience much quicker insurance processing and payouts following extreme climate events. Rapid access to these funds helps mitigate their financial vulnerabilities and promotes faster recovery for the regional agricultural economy.

To the regional and national government, this study drives the digital transformation of public agricultural services while ensuring that all sensitive records are kept strictly secure. By utilizing a secure, localized spatial database rather than vulnerable public clouds, it proves that government offices can successfully modernize disaster response, strengthen climate resilience in areas like Northern Mindanao, and improve service delivery without compromising the security and confidentiality of sensitive citizen information.

To the future researchers and developers, this project serves as a valuable technological baseline for software developers and academics interested in agricultural geoinformatics. It demonstrates a working, cost-effective framework for building secure, localized spatial platforms integrated with real-time weather data. This provides a practical foundation for future studies aiming to design innovative, non-cloud-based disaster response tools for government agencies in developing regions.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1. Preliminary Literature Review

The transition towards digital governance in agriculture relies heavily on optimizing spatial data management and mitigating environmental risks. This review examines the fundamental concepts, historical evolution, and current trends surrounding Geographic Information Systems, data centralization, and parametric agricultural insurance. The objective of this review is to synthesize existing local and international literature to establish the technical and theoretical foundations for developing a secure, localized WebGIS platform. By critically analyzing prior methodological approaches and identifying current research gaps in data privacy and disaster assessment, this review justifies the necessity of the proposed AgriSureGIS platform for the Philippine Crop Insurance Corporation Region X.

2.1.1. WebGIS and Spatial Decision Support in Agriculture

This section highlights the fundamental shift from traditional, standalone GIS to web-based platforms (WebGIS) within the agricultural sector.

The integration of GIS and internet technologies has fundamentally shifted agricultural data management towards web-based platforms. According to Patel et al. (2023), adopting open-source software such as Apache server, GeoServer, OpenLayers, and PostgreSQL with PostGIS extensions enables the development of interactive, dynamic platforms for agricultural monitoring and management. This web-based architecture allows users and decision-makers to access, visualize, and analyze spatio-temporal data over the internet without relying on monolithic, proprietary desktop software.

Handling the complex spatial and attribute data required for large-scale agricultural monitoring necessitates highly scalable system designs. To address this, Hu et al. (2024) proposed the construction of a digital agricultural GIS application suite utilizing microservice architecture and web componentization. This modular approach decomposes the system into independent service modules, allowing it to efficiently handle high concurrency and dynamic scaling while seamlessly integrating spatial data management, spatial analysis, and visualization.

Furthermore, transitioning to a WebGIS infrastructure facilitates simultaneous, multi-user collaboration, which is a significant advantage over traditional desktop GIS. Modica et al. (2025) highlighted that web-based multicriteria spatial decision support systems (MC-SDSS) allow multiple experts from various fields to interact with and evaluate geospatial data concurrently. Similarly, Avanidou et al. (2023) demonstrated the practical application of Web-GIS systems for remote farm management monitoring. By utilizing a central geodatabase linked to a web map application, their system enables land managers to remotely record, track, and analyze agricultural practices across numerous fragmented fields.

2.1.2. Eliminating Data Silos and Building Centralized Systems

This section highlights the critical challenges posed by fragmented data management, commonly referred to as “data silos”, and the operational benefits of transitioning toward centralized information architectures.

Data silos occur when information is isolated within different departments or stored in incompatible systems, which inherently hinders the free flow of information across an organization. According to Abu-AlSondos (2025), the persistence of these silos in the insurance sector causes significant delays, data inconsistencies, and suboptimal decision-making. Because different departments often rely on fragmented platforms and independent databases, organizations are forced to depend on manual data reconciliation, which is a time-consuming and error-prone process that inflates operational expenses. Furthermore, Abu-AlSondos (2025) emphasized that in highly regulated environments, the inability to generate clean, consolidated, cross-departmental data makes it practically impossible to meet strict reporting standards, thereby exposing insurers to severe compliance penalties and reputational risks.

Addressing these operational inefficiencies requires a strategic shift toward centralized computing resources and data management. Gordon (2020) highlighted that the centralization of Information and Communication Technology (ICT) offers significant advantages to organizations, including economies of scale, standardized equipment and processes, and a reduction in management overhead. Centralizing an organization's information systems ensures consistency in the data collected and significantly improves the speed and ease of aggregating data for crucial decision-making. While Gordon (2020) acknowledged the necessity of decentralized access to maintain user productivity, the core

architecture must remain centralized to effectively rationalize control and secure continuous operations.

In the context of geospatial and environmental data, eliminating silos is most effectively achieved through the implementation of Spatial Data Infrastructures (SDI). Wetzel et al. (2024) demonstrated that establishing an SDI with standardized WebGIS servers, such as GeoServer, is crucial for harmonizing fragmented data and making it interoperable across different platforms. By utilizing centralized geospatial databases and open-source web services, organizations can replace isolated data storage such as standalone desktop files with a unified system that supports seamless data exchange and concurrent multi-user access. This infrastructure allows various stakeholders to simultaneously access, search, and utilize the exact same updated spatial information, effectively dismantling previous data barriers.

2.1.3. Multi-Criteria Filtering for Spatial and Attribute Data

Effective spatial decision-making in large-scale agricultural systems depends on the ability of users to rapidly query and isolate specific data points to prevent user fatigue when browsing through thousands of records .

Cui et al. (2022) emphasize that modern WebGIS platforms must incorporate advanced filtering tools that allow users to isolate parcels based on unique features, such as land use type, field code, or individual producer IDs . For example, the FarmGeoBalance system described by Avaniadou et al. (2023) utilizes a Search and Select service that enables managers to group and view agricultural practices across numerous fragmented fields through a unified interface .

This interaction allows a Click-Feature event to select a plot and automatically highlight it on the map while simultaneously displaying its corresponding metadata such as area, coordinates, and planting date in a structured attribute box . Such multi-criteria tools are essential for GIS specialists to generate Payout-Ready summaries for management review without manually searching through voluminous datasets

2.1.4. Integration of Meteorological Data for Automated Risk Assessment

This section highlights the importance of integrating real-time meteorological data into spatial platforms to transition from manual claims verification to automated disaster risk and agricultural damage assessments.

The integration of meteorological data into spatial platforms has become a cornerstone of modern agricultural insurance and disaster risk management. Traditional claims verification processes are highly tedious and resource-intensive, but utilizing weather data models can significantly reduce these administrative burdens. According to Nguyen et al. (2025), integrating climate and weather data helps reduce basic risk and improves the design of agricultural index-based insurance. Similarly, Nguyen-Huy et al. (2024) emphasized that weather and rainfall index-based insurance can effectively mitigate financial risks for farmers exposed to extreme climate events. By integrating these meteorological variables into geoinformatic systems, agencies can rapidly estimate damages and streamline the payout process for vulnerable small-scale farmers (Rabuh et al., 2024). Furthermore, providing tailored regional climate information through Spatial Data Infrastructures (SDI) ensures that local authorities can implement evidence-based disaster responses efficiently (Wetzel et al., 2024). AgriSureGIS adopts this principle by introducing a module that automatically extracts spatial hazard data from PAGASA Tropical Cyclone Bulletins and overlays it against the PCIC Risk Exposure List. This approach not only streamlines the damage assessment workflow during typhoon events but also promotes the overall resilience of the agricultural economy (Chen et al., 2025)

2.1.5. Automated Meteorological Data Acquisition and Notification

Transitioning from manual monitoring to an automated disaster response framework requires internal scripts capable of real-time data pulling from meteorological bureaus. Wetzel et al. (2024) describe a robust implementation using Python-based scripts that regularly iterate over external data folders or websites to automatically detect and upload new contents to a WebGIS server via an API.

To maximize responsiveness, these automated workflows include notification modules such as on-screen pop-ups or automated email alerts that trigger immediately when a new report is parsed and available. This automation ensures that disaster risk assessments, which

were previously delayed by the manual downloading of files into disconnected repositories, can proceed instantly as new weather data becomes available. This streamlined process is critical for achieving the project objective of providing insurances to farmers within three to five days after a typhoon exits the Philippine Area of Responsibility (PAR).

2.1.6. Secure Localized Spatial Databases for Information Management

This section highlights the critical role of secure, localized spatial databases in centralizing large-scale agricultural records while ensuring strict data privacy.

Transitioning from fragmented data files such as standalone CSV and GPX files to a centralized spatial database is critical for handling large-scale agricultural records efficiently. According to Abu-ElSondos (2025), eliminating data silos is essential in the insurance sector, as fragmented information restricts strategic analytics and leads to delays, compliance failures, and poor decision-making. Modern spatial databases function as robust engines for geographical analysis. Specifically, object-relational database management systems like PostgreSQL, combined with its spatial extension, PostGIS, provide highly advanced features for storing, querying, and managing massive spatial datasets (Hajisheko et al., 2025; Ilba, 2021). Furthermore, utilizing localized database architectures addresses the growing necessity for strict information privacy. Shahriar et al. (2025) noted that processing large volumes of personal information poses unique privacy risks, making robust privacy-enhancing solutions imperative. By implementing a secure, localized PostGIS database instead of a public cloud repository, government agencies like the PCIC can establish strict access controls. This localized infrastructure effectively protects sensitive farmer data in compliance with data privacy regulations while simultaneously overcoming operational bottlenecks (Tissayakorn, 2025).

2.1.7. Hybrid Deployment Progressive Web Applications (PWA) and Native Frameworks

The evolution of Geographic Information Systems (GIS) deployment now favors hybrid models that ensure system resilience across varying network conditions. Ginzburg and Padalko (2020) propose the implementation of Progressive Web Applications (PWA) for GIS interfaces, highlighting their ability to provide fault-tolerant data delivery and complete

offline operation on any stationary or mobile device equipped with an HTML5-compliant browser.

This architecture utilizes advanced caching mechanisms to allow specialists to work with previously loaded spatial data even when the server is disconnected, featuring automatic synchronization of user inputs once connectivity is restored. A significant institutional benefit is the reduction in network traffic; research indicates that PWAs can save over 80% in monthly data transfer compared to traditional web applications, which is vital for regional offices with limited bandwidth.

In institutional environments like the PCIC, these web-based tools are often paired with a native "Desktop terminal" subsystem, as seen in the research by Yining et al. (2025), where a native environment is dedicated to intensive pre-field configuration, lab-based data integration, and high-level quality control before broader deployment.

2.1.8. A WebGIS Ecosystems Utilizing Google Earth Engine

Several established research frameworks demonstrate the efficacy of integrating Google Earth Engine as a backend analytical engine for WebGIS applications. Bhakal et al. (2024) developed a specialized model within the GEE framework that enables users to visualize and analyze geospatial data for a wide range of precision agriculture applications, such as assessing crop health through various indices.

In the context of disaster management, Thakur et al. (2026) utilized GEE to create an innovative framework for rapid impact assessment using Sentinel-1 Synthetic Aperture Radar data. Their system leverages the GEE API to deliver workflows that provide preliminary damage assessments within hours of new data acquisition, bypassing time-consuming manual analysis.

This integration is particularly valuable for parametric insurance because GEE can process massive multitemporal datasets to verify planted areas even during typhoon events by utilizing SAR imagery that penetrates thick cloud cover, which typically obstructs traditional optical sensors.

2.1.9. Synthesis of the Reviewed Literature

Despite these advancements, a distinct research gap exists in current literature and system development. The majority of recent agricultural WebGIS frameworks heavily emphasize either public cloud integration or mobile-offline field collection applications. There is a lack of literature addressing the specific institutional needs of government agencies that require strictly localized, non-cloud spatial databases PostGIS to comply with stringent data privacy laws, coupled with automated parametric assessment tools tailored to local meteorological bureaus.

The proposed AgriSureGIS system directly fills this gap. By utilizing an open-source FOSS architecture, it establishes a highly secure, localized platform that eliminates the need for mobile-cloud dependencies. Concurrently, by developing a Parametric Insurance Indemnification Assessment Tool that cross-references PAGASA typhoon bulletins against farm polygons, this study extends the existing knowledge of spatial decision support systems, offering a tailored, modernized workflow for disaster response in PCIC Region X.

2.2. Comparison of Existing Systems/Studies

To establish the foundation for this IT Capstone Project, a review of existing studies was conducted. Table 1 presents a comparative analysis of related works, highlighting key methodologies, system features, technological frameworks, and research gaps. This comparison provides insights into how previous studies approached similar problems and helps identify areas for improvement and innovation in the proposed system.

Feature	AgriSureGIS	CONNECTOS ED	ReKIS	WebGIS-based decision support system	WebGIS Suitability Olive	Sustainable GeoInformatic Insurance	GIS-based intelligent system for geochemical sampling	WebGIS-based platform for effective landslide hazard management
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Spatial Data Infrastructure with OGC-Compliant Web services	✓	✓	✓	✓	✓	✓	X	✓
Spatial-to-Attribute Data Linking	✓	X	X	X	X	✓	✓	✓
Parametric Insurance & Meteorological Integration	✓	X	X	X	X	✓	X	X
Automated Risk and Damage Assessment Module	✓	✓	✓	✓	✓	✓	X	✓
Advanced Location Search & Filter	✓	X	✓	✓	X	X	X	✓
Secure User Access Levels	✓	X	✓	X	✓	X	✓	✓
Interactive Visual Dashboards	✓	✓	✓	✓	✓	X	X	✓

Table 1. Comparative Analysis of Related Works

Gantt Chart

The IT Capstone Project follows a structured timeline, ensuring the completion of key milestones from February to October 2026. The project is divided into three main phases:

1. *Concept Presentation Defense (February 2026)*: Focuses on writing Chapters 1-2 and conducting initial research.
2. *Project Proposal Defense (May 2026)*: Requires the completion of the system's wireframe and revisions of Chapters 1-3.
3. *Final Defense (October 2026)*: Involves system development, testing, and documentation (Chapters 4-5).

The Gantt Chart (Figure 1) illustrates the detailed schedule and dependencies among tasks.

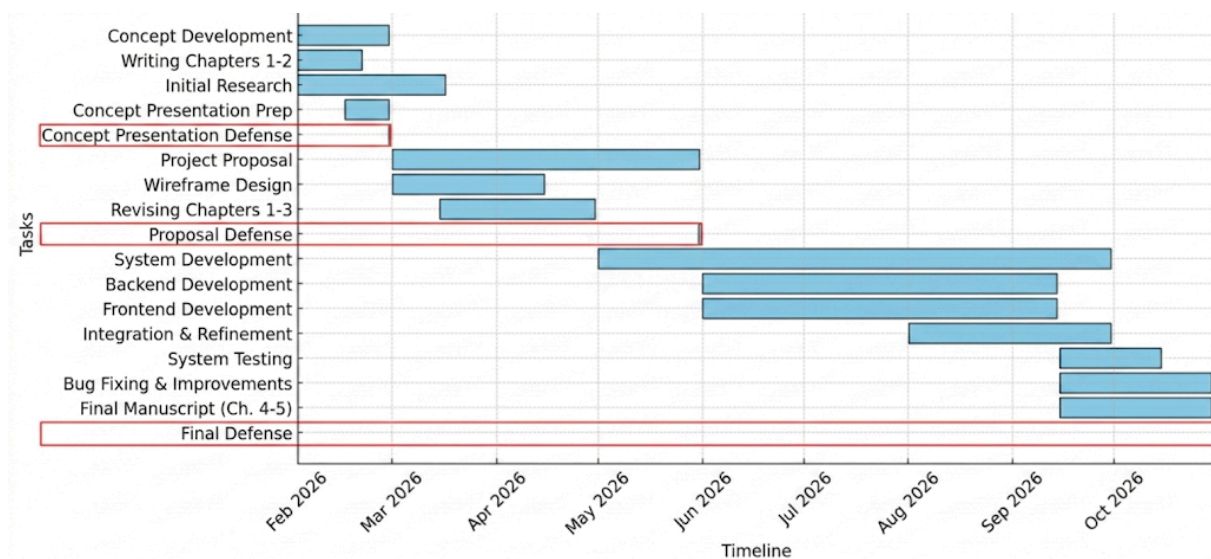


Figure 1. Gantt Chart

CHAPTER III METHODOLOGY

3.1. Software Development Model

The development of AgriSureGIS adopts the Agile Development Model, a methodology selected for its flexibility in accommodating evolving requirements through iterative and incremental cycles. This approach enables complex functionalities such as real-time PAGASA bulletin extraction and PostGIS spatial data integration to be continuously refined and optimized across successive technical sprints, rather than relying on a single deployment phase.

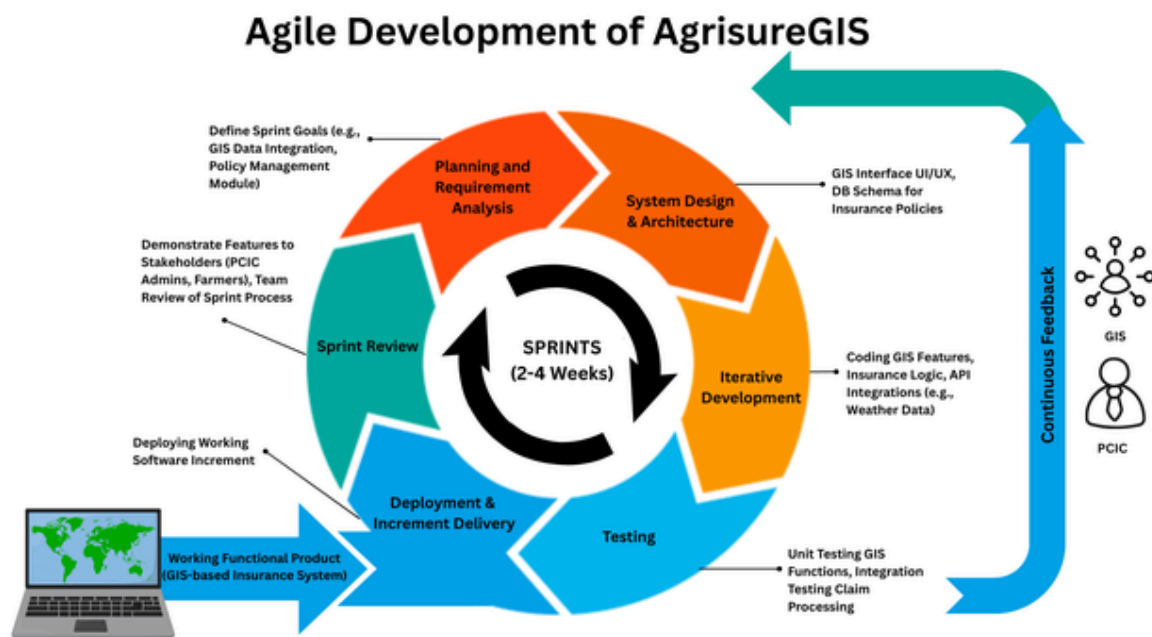


Figure 2. Agile Development Model

For the development of the AgriSureGIS platform, the proponents selected the Agile Software Development Life Cycle (SDLC), specifically utilizing a framework structured around iterative, 2-to-4 week development cycles. This methodology was deliberately chosen over rigid, sequential models (such as the Waterfall model) to address the specific operational and technical realities of building a bespoke system for a government agency.

During the initial requirements elicitation phase, it became evident that the PCIC Region X operates in a dynamic environment where reporting standards, UI preferences, and assessment rubrics may require refinement. The Agile methodology embraces changing requirements, even late in development. By breaking the project into iterative cycles, the development team can seamlessly integrate new client requests without derailing the entire project timeline.

A core principle of this project is ensuring that the final system perfectly mirrors the actual workflow of the PCIC GIS Specialists. The Agile model mandates a "Sprint Review" at the end of each cycle. This continuous feedback loop guarantees that the system is built with the client rather than just for them, effectively eliminating the risk of delivering an end-product that the agency cannot use. Moreover, the system relies heavily on advanced external integrations, specifically the Python-based scraping of PAGASA Tropical Cyclone Bulletins and the spatial radar processing via the GEE API. These are high-risk technical components. The Agile framework allows the team to isolate these complex modules into dedicated sprints.

3.2. Planning and Requirement Analysis

During this phase, the proponents engaged directly with the stakeholders of the Philippine Crop Insurance Corporation (PCIC) Region X to define the exact system boundaries, identify operational bottlenecks, and establish the technical feasibility of the proposed WebGIS solution. The team conducted a series of three comprehensive, on-site interviews at the PCIC regional office. These sessions were deliberately scheduled around the client's availability to ensure maximum participation from the primary end-users, specifically the GIS Specialists and management evaluators. The proponents also utilized synchronous virtual meetings via Google Meet ensuring that rapid clarifications could be made without disrupting the agency's daily operations. To identify the exact operational "pain points" in real-time, the proponents conducted direct observations of the agency's current procedures. PCIC management provided a controlled, discreet demonstration of their manual disaster risk assessment workflow, including an overview of the legacy PCIC Automated Business System (PABS). During this demonstration, strict data privacy protocols were enforced; all sensitive information regarding the farmers was redacted or hidden from the proponents.

Data gathering was highly iterative. Following each interview, the development team conducted internal deliberations to dissect the gathered data. The proponents mapped out the agency's existing manual disaster risk assessment workflow specifically the manual downloading of PAGASA Tropical Cyclone Bulletins (TCB) and the standalone QGIS tracing and evaluated how these processes could be translated into programmable algorithms.

3.2.1 Software Requirements Specifications

A. Functional Requirements

- The specialist shall have an account in the system.
- The specialist shall have access to all functionalities in the system.
- The system shall allow the specialist to import the existing risk exposure report (farmer records/profiles) via CSV file.
- The system shall allow the specialist to upload the farmers' spatial GPX files to establish farm polygons.
- The system shall allow the specialist to execute the automated analysis, triggering the system to generate the Tropical Cyclone Bulletin (TCB) data, which includes period of exposure and wind velocity.
- The system shall automate the calculation process by mapping the period of exposure, identifying the wind velocity, matching the crop growth stage, and calculating the corresponding indemnity/damage factor.
- The system shall allow the specialist to filter the dashboard and reports by municipality, province, barangay, or city.
- The system shall allow the specialist to sort records alphabetically, by planting date, or by any specific column.
- The system shall allow the specialist to generate and export an extended CSV or Excel report that appends the calculated results (period of exposure, wind velocity, indemnity factor, and final indemnity payment) to the original PCIC row format.
- The system shall allow the administrator to securely log in to access the backend infrastructure.

- The system shall allow the administrator to create or restrict user accounts, manage localized system configurations, and execute database backups to ensure institutional data security.
- The system shall provide a dedicated "Calibration/Settings" module on the dashboard for configuring system parameters.
- The system shall allow the administrator to input or manage Google Cloud Project credentials to allocate the capacity for Google Earth Engine SAR analysis.
- The system's Python-based parser shall monitor the PAGASA website and automatically download the Tropical Cyclone Bulletin updates during an active typhoon in the PAR.
- The system shall trigger an on-screen pop-up and an automated email alert to notify the GIS specialist when a new bulletin is successfully parsed.
- The system shall automatically compile and store all downloaded versions of the PAGASA PDF bulletins during a typhoon event without overwriting previous versions.
- The system shall interface with Google Earth Engine to retrieve Synthetic Aperture Radar (SAR) imagery to analyze if the area is planted within the polygons.
- The main dashboard shall be structurally divided into four specific modules to handle the automated disaster risk assessment workflow: a Home Dashboard, Spatial Analysis & Data Import, Assessment & Reporting, and Calibration/Settings.
- The system shall provide a drag-and-drop area for uploading new CSV files.
- The system shall display corresponding metadata (farmer ID, name, address, farm ID, hectare area, and whether it is planted) instantly when a user clicks on a farm polygon on the map.
- The system shall display warning prompts before a user deletes, overwrites, or finalizes any data.
- The system shall provide a "Generate Report" button that allows users to preview the CSV layout before finalizing the download
- The system shall utilize specific UI theme colors combining green, yellow (gold), and blue, and support both light and dark modes.

B. Non-Functional Requirements

- **Accessibility** - The web interface shall be dynamically responsive so that it is readable and functional on varying screen sizes, including smaller monitors used during field office meetings.
- **Accuracy** - The system must accurately extract variables such as time, Tropical Cyclone Wind Signal number, wind velocity, and affected areas directly from PAGASA's PDF bulletins.
- **Compatibility** - The system shall be compatible with and accessible across multiple web browsers.
- **Performance** - The system shall run heavy geoprocessing calculations in the background while displaying a loading spinner to the user.
- **Usability** - The graphical interface must be simple, functional, and easy to navigate, integrating PCIC's official theme colors of green, yellow (gold), and blue.
- The interface must support both light and dark modes.
- The system must prompt the user with warning messages before allowing them to delete, overwrite, or finalize data.
- **Reliability** - The system must automatically save ongoing work and imported data.
- **Security** - Spatial boundaries and sensitive insurance data must be securely hosted on a localized PostgreSQL database with the PostGIS extension, specifically avoiding public cloud servers.
- The system must retain the last active session to prevent data loss, in the event of a session timeout due to 5 minutes of inactivity.
- **Data Consistency** - The system must generate consistent CSV files that the PCIC finance department can utilize without manual editing, appending the calculation columns to the end of the existing PCIC format.
- **Fault Tolerance** - If the system fails to parse the PAGASA PDF bulletins, it must explicitly alert the user with specific error messages rather than crashing.
- The system must be designed to retain baseline functionality even if the PAGASA website goes down or experiences an outage.

C. System Modules

- **Monitoring and Extraction Module:** This module serves as the landing page and active monitoring center for the system. It handles the real-time pulling and extraction of PAGASA Tropical Cyclone Bulletin (TCB) reports and features a notification system to immediately alert the GIS specialist when a new TCB is successfully parsed and downloaded.
- **Spatial Analysis and Data Import Module:** This module maximizes visualization and data handling by dividing the interface into a top and bottom layout. The top panel features the interactive GIS base map that displays the overlay of the typhoon's trajectory and the mapped polygons of the affected insured rice farms, while also allowing the export of the initial "Period of Exposure" report. The bottom panel provides a drag-and-drop interface for importing the legacy farmer record CSV files and spatial GPX polygons, displaying a tabular overview where clicking a row automatically highlights the corresponding farm polygon on the map above.
- **Assessment and Reporting Module:** This module serves as the data compilation and review center for indemnification processing. It presents a summarized view of the parsed TCB records alongside the imported rice risk exposure records, automates the extended tabular calculation results (factoring in the period of exposure, wind velocity, crop growth stage, and indemnity factors), and features a "Generate Report" button to preview and export the finalized CSV indemnification report.
- **Calibration and Settings Module:** This module functions as the administrative and configuration panel, accessed via a toggle menu to maximize the vertical screen space for the map interface. It allows authorized users to manually edit system parameters, or override data as a fail-safe, and manage system credentials such as the Google Cloud Project IDs used to allocate capacity for Google Earth Engine satellite analysis.

3.3. System Design

3.3.1 System Architecture

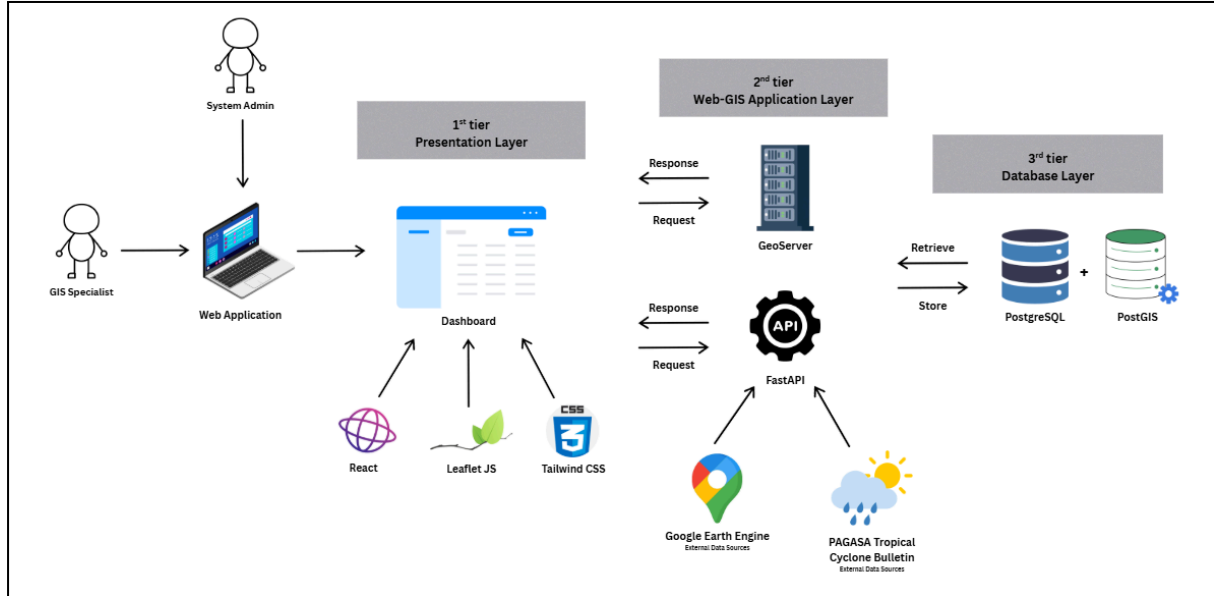


Figure 3. System Architecture

As shown in Figure 3, the proposed system utilizes a robust three-tier architecture tailored for spatial data processing and web-based visualization. The first tier, or Presentation Layer, provides the user interface for the GIS Specialist and System Administrator through a dynamic web dashboard built with React, styled with Tailwind CSS, and integrated with Leaflet JS for interactive map rendering. The second tier, the Web-GIS Application Layer, acts as the system's processing core; it employs FastAPI to manage API routing and fetch external meteorological and satellite data from the PAGASA Tropical Cyclone Bulletin and Google Earth Engine, while GeoServer operates as the dedicated spatial engine to process and serve geographic data. Finally, the third tier, the Database Layer, ensures secure and persistent storage by leveraging PostgreSQL for structured relational data alongside the PostGIS extension for advanced spatial querying and geospatial data management.

3.3.2 Top Level Diagram

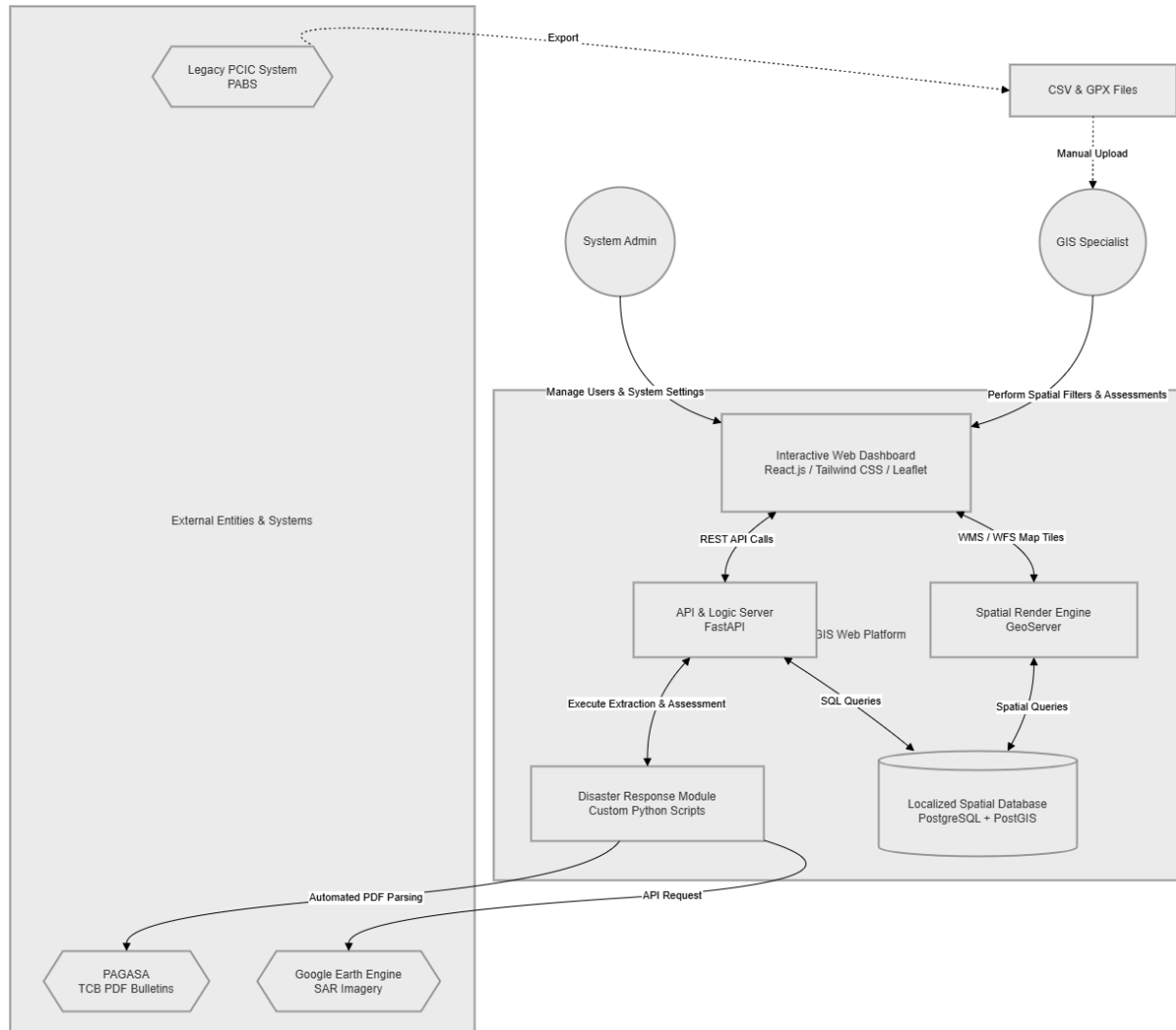


Figure 4. Top Level Diagram

As shown in figure 4, the primary client interface, engineered as an Interactive Web Dashboard utilizing React.js, Tailwind CSS, and Leaflet, provides responsive user viewports and map layouts that handle client-side rendering and state transitions dynamically without full-page reloads. This presentation layer isolates its operational dependencies by channeling administrative data requests and transaction states to the central API and Logic Server driven by FastAPI. This backend orchestrator capitalizes on native asynchronous processing loops to handle concurrent user requests, route validation, and administrative workflows without systemic latency.

Heavy-duty data processing and geographic file manipulation are systematically delegated to a specialized Disaster Response Module powered by custom Python scripts. This pipeline automates the extraction, transformation, and loading (ETL) required to parse zipped coordinates and tabular matrices, transforming raw inputs into structured datasets while managing programmatic API queries to Google Earth Engine and automated file fetching from PAGASA nodes.

To optimize visualization performance and eliminate client-side computing bottlenecks, spatial rendering is decoupled from primary application logic and assigned to a dedicated GeoServer instance. GeoServer monitors the database layer and streams map data onto the frontend Leaflet canvas using standardized, high-performance Open Geospatial Consortium (OGC) Web Map Service (WMS) and Web Feature Service (WFS) protocols.

Data persistence and geographic computing are maintained within a Localized Spatial Database instance running a PostgreSQL relational engine extended by the PostGIS spatial database framework. This persistent storage tier houses standard alphanumeric user profiles alongside complex geometric vectors like farm polygons and storm trajectories. By leveraging specialized spatial SQL operators and Generalized Search Tree (GiST) indexing, the database layer resolves multi-conditional parametric evaluations and proximity calculations natively at the storage tier, ensuring relational integrity and enterprise scalability across all platform components.

3.3.2 Use Case Diagram

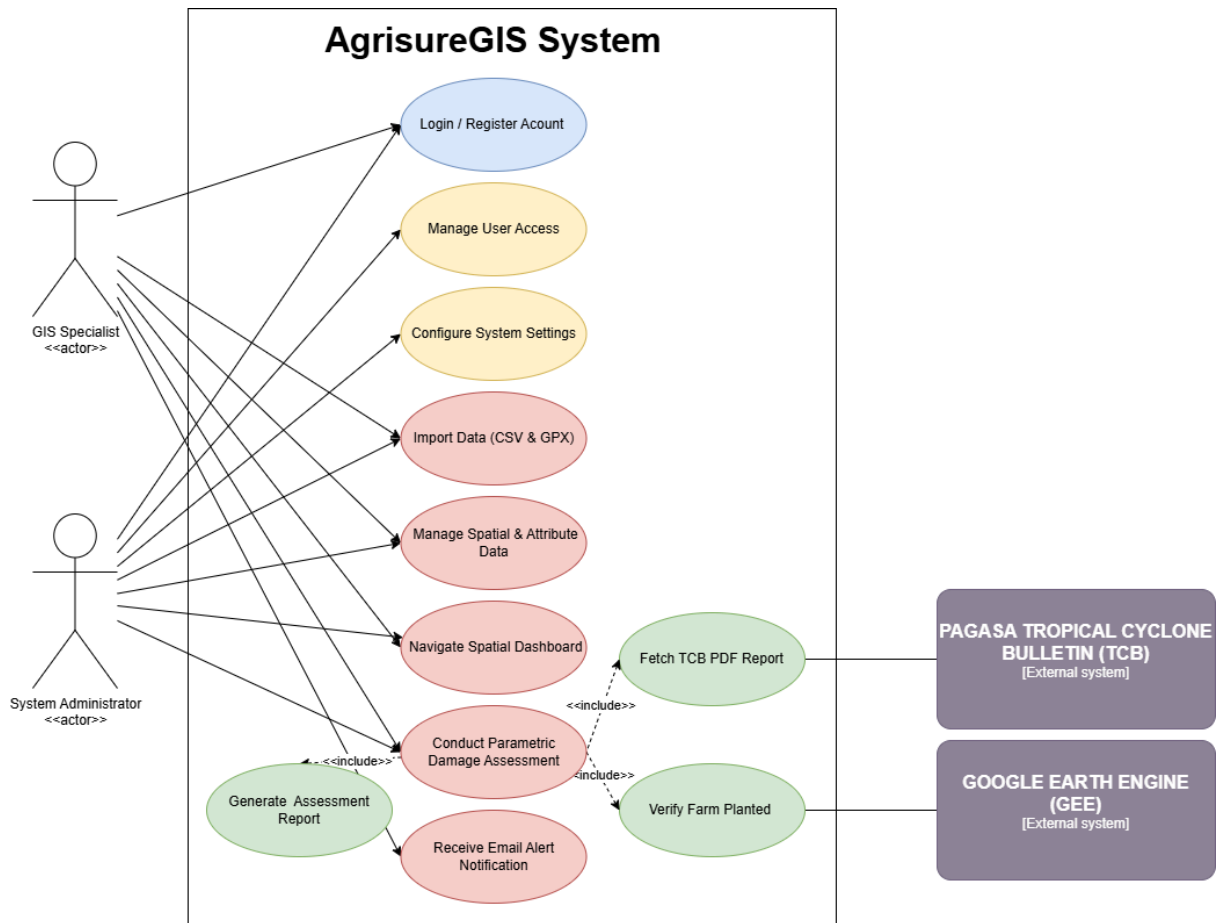


Figure 5. Use Case Diagram

The system's use case is shown in figure 5, illustrating the interactions between two primary human actors and two external system entities required to execute the agricultural risk assessment workflows. The internal actors include the GIS Specialist, who serves as the primary operational user tasked with managing spatial and attribute data including the importation of Risk Exposure Lists and GPX files, navigating the spatial dashboard, conducting parametric damage assessments, receiving email alert notifications and generating final reports; and the System Administrator, who oversees system governance, account authentication, and user access. To support these core functions, the system integrates with secondary external entities. Specifically, it employs an include relationship with the PAGASA Tropical Cyclone Bulletin to fetch essential meteorological reports during an assessment, and a relationship with Google Earth Engine to conditionally verify planted farm statuses using remote sensing data.

3.3.3 Context Diagram

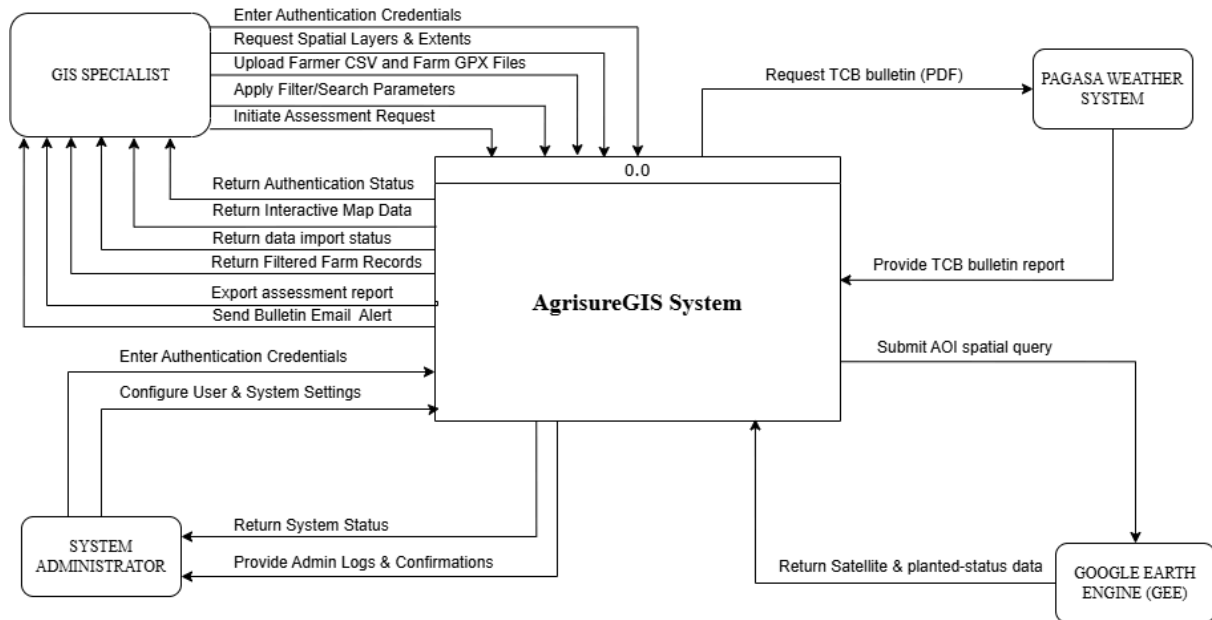
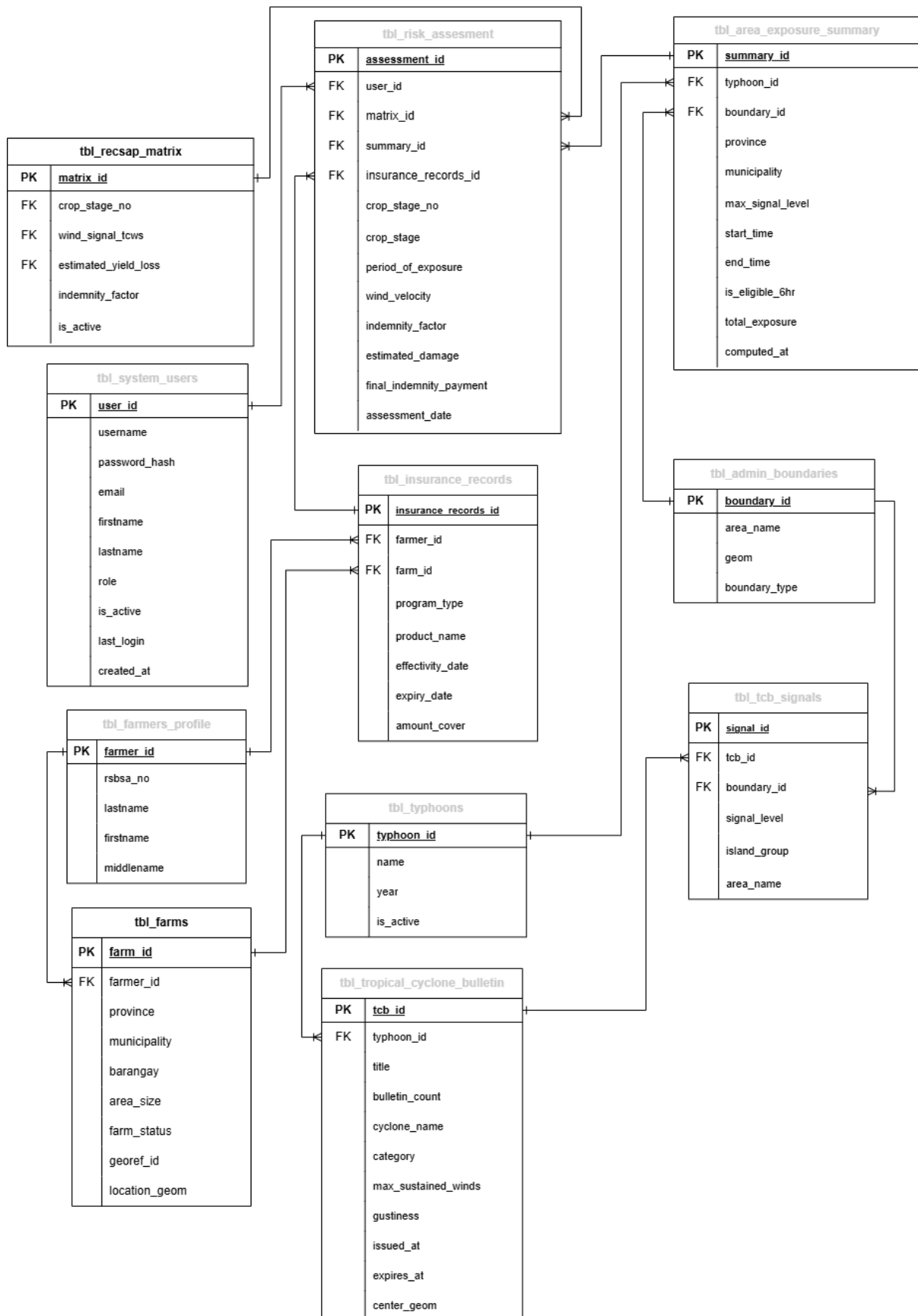


Figure 6. Context Diagram

At the operational core shown in Figure 6, is the GIS Specialist, who supplies the system with critical inputs by entering authentication credentials, requesting spatial layers and extents, uploading farmer CSV and farm GPX files, applying filter and search parameters, and initiating assessment requests. In response, the system processes these interactions and outputs the authentication status, interactive map data, data import status, filtered farm records, exported assessment reports, and bulletin email alerts directly back to the user. System governance is maintained by the System Administrator, who interacts with the central system by entering authentication credentials and configuring user and system settings. The system subsequently returns the overall system status, alongside administrative logs and confirmations, to this actor. To execute its specialized spatial and parametric analyses, the AgrisureGIS System automatically interfaces with secondary external entities. It actively sends requests for TCB bulletins (PDF) to the PAGASA Weather System, which in turn provides the necessary TCB bulletin reports. Concurrently, the system submits Area of Interest (AOI) spatial queries to Google Earth Engine, which responds by returning crucial satellite imagery and planted-status data.

3.3.4 Data Flow Diagram

3.3.5 Entity Relationship Diagram



3.3.5.1 Database Dictionary

Table: tbl_system_users				
Field Name	Data Type	Length	Constraint	Description
user_id	SERIAL	50	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the staff member.
username	VARCHAR	50	UNIQUE, NOT NULL	The username used to log into the platform.
password_hash	VARCHAR	255	NOT NULL	The cryptographically hashed the password.
email	VARCHAR	100	UNIQUE, NOT NULL	Staff member's official email address.
firstname	VARCHAR	100	NOT NULL	Staff member's first name.
lastname	VARCHAR	100	NOT NULL	Staff member's last name.
role	ENUM	50	NOT NULL	Defines their permission level.
is_active	BOOLEAN	-	NOT NULL, DEFAULT TRUE	Toggle to easily suspend an account without deleting their data.
last_login	TIMESTAMPTZ	-	NULLABLE	Timestamp of the user's most recent successful login.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the account was created.
Table: tbl_farmers_profile				
Field Name	Data Type	Length	Constraint	Description
farmer_id	INT	50	PRIMARY KEY	Unique identifier for the farmer.
rsbsa_no	VARCHAR	50	UNIQUE, NULLABLE	Registry System for Basic Sectors in Agriculture number.

lastname	VARCHAR	100	NOT NULL	Farmer's last name.
firstname	VARCHAR	100	NOT NULL	Farmer's first name.
middlename	VARCHAR	100	NULLABLE	Farmer's middle name.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farmer profile was created.
updated_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farmer profile was updated.
Table: tbl_farms				
Field Name	Data Type	Length	Constraint	Description
farm_id	INT	50	PRIMARY KEY	Unique identifier for the land parcel.
farmer_id	INT	50	FOREIGN KEY	References farmers.farmer_id. Connects land to the owner.
province	VARCHAR	100	NOT NULL	Regional location of the farm.
municipality	VARCHAR	100	NOT NULL	City or town location of the farm.
barangay	VARCHAR	100	NOT NULL	Local neighborhood/village of the farm.
area_size	DECIMAL	10, 4	NOT NULL	Size of the land in hectares or square meters.
Is_planted	BOOLEAN	-	NOT NULL, DEFAULT FALSE	Binary flag verifying if the farm is actually planted, derived via SAR imagery.
georef_id	VARCHAR	100	NULLABLE	The specific map code used to link this record to your spatial files (e.g., "R10-13-16-012-0000 16").
location_geom	GEOMETRY	GEOM	NULLABLE	PostGIS spatial data

		ENTRY(MultiPolygon, 4326)		representing the exact boundary of the farm. Populated via spatial join with the georef_id.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farm record was created.
updated_at	TIMESTAMPTZ		NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farm record was updated.
Table: tbl_insurance_record				
Field Name	Data Type	Length	Constraint	Description
insurance_record_id	SERIAL	50	PRIMARY KEY	Unique identifier for the insurance contract.
farmer_id	INT	50	FOREIGN KEY	References tbl_farmers_profile.farmer_id. Connects the farmer to the insurance record.
farm_id	INT	50	FOREIGN KEY	References farms.farm_id. Connects the policy to the land.
program_type	VARCHAR	100	NOT NULL	Type of insurance program.
product_name	VARCHAR	150	NOT NULL	Specific name of the insurance product.
effectivity_date	DATE	-	NOT NULL	Start date of the insurance coverage.
expiry_date	DATE	-	NOT NULL	End date of the insurance coverage.
amount_cover	DECIMAL	15, 2	NOT NULL	The maximum financial payout amount.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the insurance record was created.

updated_at	TIMESTAMPTZ		NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the insurance record was updated.
Table: tbl_risk_assessments				
Field Name	Data Type	Length	Constraint	Description
assessment_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the report.
insurance_record_id	INT	50	FOREIGN KEY	References tbl_insurance_record.insurance_records_id. Links report to the contract.
summary_id	INT	4	FOREIGN KEY	References tbl_area_exposure_summary.summary_id. Links.
matrix_id	INT		FOREIGN KEY	References tbl_recsap_matrix.matrix_id.
crop_stage_no	INT		NOT NULL	Numerical indicator of the crop's growth stage.
crop_stage	VARCHAR	150	NOT NULL	Text description of the growth stage
period_of_exposure	INT		NOT NULL	The calculated age of the crop in days at the time of the typhoon, derived by subtracting the effectivity_date from the typhoon occurrence date.
wind_velocity	INT		NOT NULL	The exact wind speed (in km/h) that the crop was subjected to.
indemnity_factor	DECIMAL	5, 4		The multiplier pulled from the look-up table based on crop type, stage, and wind

				velocity.
estimated_damage	DECIMAL	15, 2	NOT NULL	The calculated financial loss or damage estimate.
final_indemnity_payment	DECIMAL	15, 2	NOT NULL	The final calculated monetary payout required for the Finance Division report (amount_cover * indemnity_factor).
assessment_date	TIMESTAMPTZ	-	NOT NULL	Necessary for tracking when the calculation/assessment was performed, ensuring accurate record-keeping for management review and payout approval
user_id	INT		FOREIGN KEY	Necessary for tracking user ensuring accurate record-keeping for user.

Table: tbl_typhoons

Field Name	Data Type	Length	Constraint	Description
typhoon_id	SERIAL		AUTO_INCREMENT, PRIMARY KEY	Internal auto-incrementing unique ID utilized for system-level database joins and relational data management.
name	VARCHAR	100	NOT NULL	The assigned name of the storm (e.g., "Leon").
year	INT	4	NOT NULL	The year the storm occurred (e.g., 2026), preventing issues when PAGASA recycles names.
is_active	BOOLEAN		NOT NULL, DEFAULT	Master switch. When

			TRUE	the storm leaves the PAR, this flips to FALSE and triggers the final payout computation.
Table: tbl_tropical_cyclone_bulletins				
Field Name	Data Type	Length	Constraint	Description
tcb_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the bulletin.
typhoon_id	INT		FOREIGN KEY	References tbl_typhoons.typhoon_id for the bulletin.
title	VARCHAR	255	NOT NULL	Full official title of the bulletin.
bulletin_count	INT	10	NOT NULL	the sequential number of the bulletin (e.g., 1, 2, 3).
category	VARCHAR	100	NOT NULL	Storm classification (e.g., "Tropical Depression").
max_sustained_winds	INT	10	NULLABLE	The actual maximum sustained wind speed near the center (typically stored as km/h).
gustiness	INT	10	NOT NULL	The actual maximum wind gusts (typically stored as km/h).
issued_at	TIMESTAMPTZ		NOT NULL	Timestamp when the bulletin was officially published.
expires_at	TIMESTAMPTZ		NOT NULL	Timestamp when the current bulletin is no longer valid.
center_geom	geometry	Point, 4326	NOT NULL	PostGIS spatial data representing the storm's eye/center.
Table: tbl_tcb_signals				

Field Name	Data Type	Length	Constraint	Description
signal_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the signal record.
tcb_id	INTEGER	4	FOREIGN KEY	References tbl_tropical_cyclone_b ulletins.tcb_id. Links warning of the storm.
signal_level	INTEGER	4	NOT NULL	The TCWS warning level (e.g., 1, 2, 3, 4, 5).
island_group	INTEGER	4	NOT NULL	Regional grouping code (0 = Luzon, 1 = Visayas, 2 = Mindanao).
area_name	VARCHAR	100	NOT NULL	The text name of the affected province/island.

Table: tbl_area_exposure_summary

Field Name	Data Type	Length	Constraint	Description
summary_id	SERIAL		AUTO_INCREMENT, PRIMARY KEY	Unique identifier for this specific area-storm summary.
typhoon_id	INTEGER		FOREIGN KEY	References tbl_typhoons.typhoon _id. Identifies which storm caused this exposure.
province	VARCHAR	100	NOT NULL	The administrative province name (e.g., "Misamis Oriental").
municipality	VARCHAR	100	NOT NULL	The specific municipality or city name (e.g., "Talakag").
max_signal_level	INTEGER	1	NOT NULL	The highest (peak) TCWS level recorded for this area during the storm.

start_time	TIMESTAMPTZ		NOT NULL	The issued_at timestamp of the first bulletin that placed this area under a signal.
end_time	TIMESTAMPTZ		NOT NULL	The expires_at timestamp of the last bulletin that included this area.
is_eligible_6hr	BOOLEAN		DEFAULT FALSE	The Trigger: Automatically set to TRUE if max_signal_level >= 2 AND total_exposure_hours >= 6.0.
total_exposure_hours	DECIMAL	5, 2	NOT NULL	The mathematical sum of all active signal windows ($\sum$ expires - issued).
computed_at	TIMESTAMPTZ		DEFAULT NOW()	Timestamp of when this summary was generated by the backend.

Table: tbl_admin_boundaries

Field Name	Data Type	Length	Constraint	Description
boundary_id	SERIAL	-	PRIMARY KEY	Auto-incrementing unique ID for the map feature.
psgc_code	VARCHAR	10	UNIQUE, NOT NULL	The official 10-digit Philippine Standard Geographic Code. Prevents duplicate entries and spelling errors.

area_name	VARCHAR	100	NOT NULL, PARTIAL UNIQUE	Standardized name. Constrained by a Partial Unique Index (idx_unique_province_na me) to enforce uniqueness ONLY when boundary_type = 'Province'.
boundary_type	ENUM	-	NOT NULL	Classification level. Must be: 'Region', 'Province', or 'Municipality'.
parent_boundar y_id	INT	-	FOREIGN KEY	Self-referencing link to boundary_id. Links a Municipality to its Province. Left NULL if the record is a Region or Province.
geom	GEOMETRY	MultiPol ygon, 4326	NOT NULL	PostGIS spatial data representing the area's official administrative boundary.

Table: tbl_recsap_matrix

Field Name	Data Type	Length	Constraint	Description
matrix_id	SERIAL	-	AUTO_INCREME NT, PRIMARY KEY	Unique identifier for this specific parametric rule.
crop_stage_n o	INT	-	NOT NULL	Numerical indicator of the crop's growth stage (0 through 9).
wind_signal_t cws	INT	-	NOT NULL	The PAGASA Tropical Cyclone Wind Signal level (1 through 5).
estimated_yie ld_loss	DECIMAL	5, 2	NOT NULL	Standard expected loss

				percentage (e.g., 20.00, 30.00) for UI display.
indemnity_factor	DECIMAL	5, 4	NOT NULL	The strict multiplier (e.g., 0.2640, 0.3920, 1.0000) used by the backend to compute the final payout.
is_active	BOOLEAN	-	NOT NULL, DEFAULT TRUE	Allows PCIC to update rates in the future without breaking historical claims.

Table 2. Database Dictionary

3.4. System Development

3.4.1 Hardware Requirements

Hardware Component	Desktop Minimum	Desktop Recommended	Laptop Minimum	Laptop Recommended
Processor (CPU)	Intel Core i5 (11th Gen) or AMD Ryzen 5	Intel Core i7 (12th Gen) or AMD Ryzen 7	Intel Core i5 (11th Gen) or AMD Ryzen 5	Intel Core i7 (12th Gen) or AMD Ryzen 7
Memory (RAM)	16 GB DDR4	32 GB DDR4/DDR5	16 GB DDR4	32 GB DDR4/DDR5
Storage	256 GB SSD	512 GB to 1 TB SSD	256 GB SSD	512 GB to 1 TB SSD
Graphics (GPU)	Integrated Intel Iris Xe / AMD Radeon Graphics	Dedicated NVIDIA RTX 3050 / 4050 Mobile	Integrated Intel Iris Xe / AMD Radeon Graphics	Dedicated NVIDIA RTX 3050 / 4050 Mobile

Table 3. Hardware Requirements

To effectively and efficiently develop and deploy AgriSureGIS, the hardware framework was established based on the desktop infrastructure used by PCIC GIS specialists and the laptops used by the proponents. This setup provides the minimum baseline and recommended upgrades required to run the web-based GIS application, host the localized database, integrate Google Earth Engine, and render massive farm polygon datasets without performance bottlenecks.

3.4.2 Software Requirements

The AgrisureGIS system will utilize a modern and scalable software stack designed to support web-based geospatial map visualization, automated parametric crop damage assessment, remote sensing-based farm verification, automated meteorological data scraping, and administrative user management. The system will include dynamic web-based dashboard functionalities to provide GIS specialists and system administrators with a unified and efficient agricultural risk assessment and insurance verification platform.

To support the modular, three-tier architecture of AgrisureGIS, the following software tools and frameworks will be utilized:

Web Application Development

The core development of the AgrisureGIS web interface is engineered utilizing React and Tailwind CSS to deliver a highly modular and responsive user experience. React serves as the foundational component-based frontend library, managing client-side state transitions dynamically to allow real-time updates and seamless interactivity without full-page reloads. Tailwind CSS is fully integrated into this presentation layer, leveraging its utility-first design classes to construct a fluid, cross-browser responsive layout optimized for data-heavy administrative layouts.

Web-Based Spatial Dashboard

The interactive, map-centric capabilities of the platform are driven by the integration of Leaflet JS and GeoServer. Leaflet JS functions as the primary client-side mapping engine, embedded within the frontend architecture to render interactive map viewports, handle base imagery layers, and dynamically map clickable farm vector data. Bridging the gap between storage and display, GeoServer acts as the dedicated map server, establishing direct connections to the spatial database to translate geographic attributes and stream them onto the Leaflet canvas using standardized, high-performance Web Map Service (WMS) and Web Feature Service (WFS) protocols.

File Ingestion and Extract, Transform, Load (ETL) Pipeline

To accommodate the importation of compressed or loose multi-sourced files from the PCIC, the application layer utilizes a specialized polymorphic automation script pipeline written in Python and executed within FastAPI. File extraction, workspace mapping, and system background decompression are driven natively by the integrated utilization of the Python zipfile and pathlib standard libraries, while boundary duplicate checking is managed through cryptographic validation filters powered by hashlib to verify payload uniqueness before deep record processing. Alphanumeric record transformation is achieved by integrating the pandas data manipulation framework to parse, clean, and isolate structural attributes from unzipped .csv registries. In parallel, geometric structures are constructed by parsing raw Extensible Markup Language (XML) data streams from field-captured .gpx tracks using the gpxpy extraction library.

To ensure topological validity, these separate spatial indicators are mapped within the GeoPandas framework, where low-level geometric engines driven by Shapely enforce strict validation rules to weld individual waypoints into valid, uncorrupted geographic boundaries (POLYGON or MULTIPOLYGON features). To align diverse coordinates with web mapping layers and weather footprints, a coordinate transformation component driven by pyproj performs mathematical projection mapping, standardizing all spatial layers to the global WGS 84 (SRID 4326) reference datum. Data persistence and Object-Relational Mapping (ORM) transactions are maintained by converting the parsed GPX streams directly into native geometry types, leveraging SQLAlchemy to stream alphanumeric metadata records alongside GeoAlchemy2 to map and insert these spatial features directly into persistent PostgreSQL/PostGIS geometry columns.

Backend and API Services

text

Development and Environment Tools

3.5. System Testing and Evaluation

3.5.1 Functionality Testing

Functionality testing will be conducted to verify the system's ability to accurately ingest, process, and clean incoming data packages under a unified backend workflow. Testing will verify the platform's capability to intercept loose multi-file submissions of tabular records and geographic coordinates, alongside bulk compressed archives, ensuring both entry points normalize data identically. Perimeter security checks will verify the application's ability to calculate cryptographic content hashes, confirming that duplicate user files are caught and rejected before triggering database transactions.

Additional validation testing will be conducted to verify the system's ability to execute cross-runtime tasks when capturing real-time meteorological indicators. By injecting mock weather bulletins with known text parameters, testing will confirm that pattern-matching scripts parse: cyclone metadata, local wind risk signals, and forecast coordinates into structured data matrices and are extracted with exact precision.

Geospatial processing audits will be conducted to verify the system's ability to convert raw global positioning coordinates directly into spatial database geometries standardized to a uniform global datum. This phase will evaluate execution rules to confirm that automated hazard buffers generated along calculated storm paths successfully intersect target farm boundaries without data loss. Finally, structural resilience testing will be conducted to verify the system's ability to handle runtime anomalies, ensuring that unexpected script errors or API network drops instantly trigger database rollbacks and storage purges to protect baseline server memory.

3.5.2 Parametric Calculation and Spatial Accuracy Testing

Evaluating the computational precision of the disaster assessment engine requires strict validation of the algorithmic logic and spatial join operations that determine insurance payouts. Algorithmic compliance testing verifies that the system correctly translates complex relational logic rules natively at the database level by cross-referencing localized crop growth stages against meteorological wind signals to isolate exact payout percentages from reference matrices.

To ensure mathematical precision during large-scale automated assessments, spatial intersection audits test the processing layer against known control datasets. This validation method uses fixed geometric overlaps and simulated buffer shapes to check that the calculated loss area matches exact geographic realities, ensuring that overlapping spatial features or topological edge cases do not trigger calculation errors or misaligned risk ratings. Furthermore, data flow tracking exercises monitor the movement of values from the initial hazard calculation through to final database logging. This verifies that individual farmer records, audited loss percentages, and finalized financial multipliers map cleanly across relational database tables without discrepancies, guaranteeing mathematical accuracy across all processing cycles.

3.5.3 Operational Usability Evaluation

Usability evaluations will be conducted to verify the system's ability to satisfy the operational demands of authorized institutional personnel during real-world simulation runs. Task-oriented user walkthroughs will verify the platform's ability to support seamless core business workflows, tracking whether the specialist can successfully execute bulk file uploads, monitor live progress indicators, trigger automated risk assessments against active storm prints, and download final audit sheets without technical friction. Compliance reviews will be conducted to verify the system's ability to generate high-quality, formatted report sheets directly from the web interface. Authorized personnel will audit exported spreadsheets to confirm that programmatically injected indemnity values, system-calculated loss parameters, and layout treatments completely match strict corporate reporting criteria and institutional guidelines, verifying that the final tool effectively achieves rapid disaster risk assessments.

3.6. Wireframe

3.6.1 Low Fidelity Wireframe

Sample 2 Low fidelity


Automated Disaster Risk Assessment System

Hello username! 

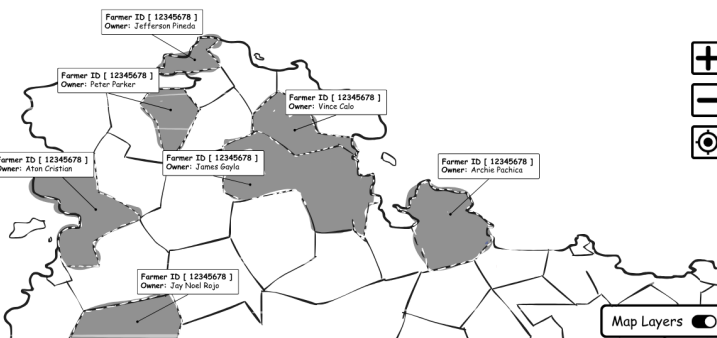
Interactive GIS Base Map

☒ Signal No. 1

☒ Signal No. 2

☒ Signal No. 3

☒ Signal No. 4



Farmer ID [12345678]
Owner: Jefferson Pineda

Farmer ID [12345678]
Owner: Peter Parker

Farmer ID [12345678]
Owner: Vince Cole

Farmer ID [12345678]
Owner: James Gayle

Farmer ID [12345678]
Owner: Archie Pothica

Farmer ID [12345678]
Owner: Alan Crastan

Farmer ID [12345678]
Owner: Jay Noel Raja

+

-



Map Layers ☐

Data Import & Preview

Farmer ID	Name	Farm ID	Planted(SAR)	Hectares	CSV/GPX Status
					


Drop CSV/GPX Files here to import

Low Fidelity 1. Spatial Analysis and Data Import Module


Automated Disaster Risk Assessment System

Hello username! 

Real Time Feed

Active TCB Monitoring [Typhoon Name - Active]

PAGASA Bulletin Status

Ok / Latest Bulletin
[Time] Parsed



[Time]

PAGASA Bulletin Status

Ok / Latest Bulletin
[Time] Parsed



[Time]

Notification 

Recent Alerts

☒ New TCB Parsed: [Time]

☒ New TCB Parsed: [Time]

☒ New TCB Parsed: [Time]

☒ New TCB Parsed: [Time]

☒ New TCB Parsed: [Time]

PAGASA Bulletin Archives

[Download PDF]

Low Fidelity 2. Monitoring and Extraction Module

45

Automated Disaster Risk Assessment System

Hello username! 

Filters

[Municipality] ▼

[Province] ▼

[Barrangay] ▼

[Crop Stage] ▼

Farmer ID	Farm ID	TCB Bulletin Time	TCB Wind (m/s)	Signal	Period of Exposure (hrs)	Crop Stage	Indemnity Damage Factor	Final Indemnity Payout

Run Automated Analysis

Preview CSV Report

Low Fidelity 3. Assessment and Reporting Module

System

Parameters

Notification

text

Verification

Calibration Settings

Sytem Timeout: [5 min]

Notification Setting ☒

Fail-safe Override ☐

Google Earth Engine Integration ☒

Google Cloud Project Credentials []

Save

Low Fidelity 4. Calibration and Settings Module

3.6.2 High Fidelity Wireframe

3.7. Deployment

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